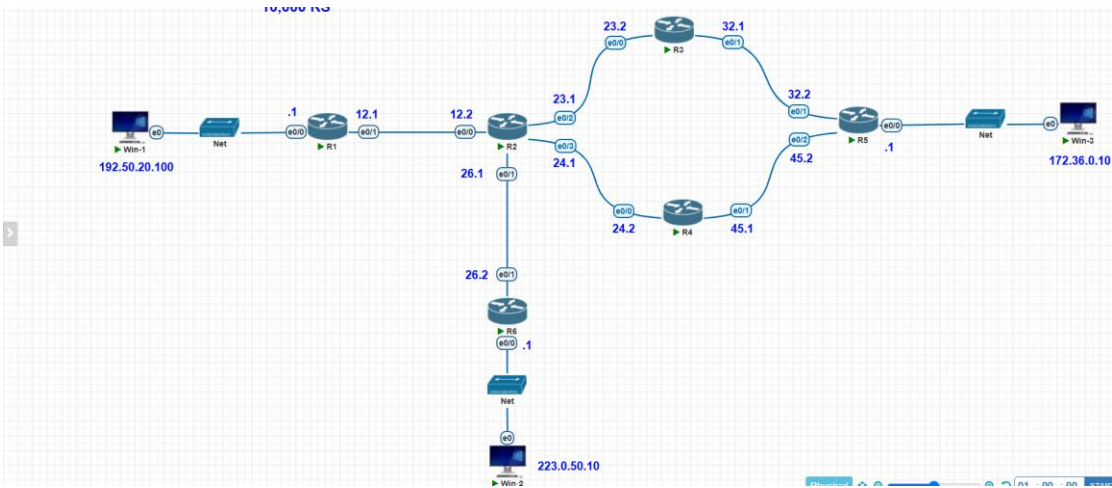


Project -2: Ring Topology Static Route Table



Ping:Win1 to Win2,Win3|Ping:Win 2 to Win1,Win3|Ping:Win3 to Win1,Win2:

```

UltraVNC Viewer - Connection dropped, trying to reconnect (3)
C:\WINDOWS\system32\cmd.exe
C:\WINDOWS\system32\cmd.exe
C:\WINDOWS\system32\cmd.exe

Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\User>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\User>

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    IP Address. . . . . : 192.50.20.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.50.20.1

C:\Documents and Settings\User>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\User>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\User>

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    IP Address. . . . . : 172.36.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 172.36.0.1

C:\Documents and Settings\User>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\User>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\User>
    
```

R1,R2: Route Table

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/1
L    12.0.0.1/32 is directly connected, Ethernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
C    192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.50.20.0/24 is directly connected, Ethernet0/0
L    192.50.20.1/32 is directly connected, Ethernet0/0
S    223.0.50.0/24 [1/0] via 12.0.0.2

R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
S    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
C    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
C    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 24.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
    
```

R3,R4: Route Table

```
R4
Router#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    24.0.0.2        YES manual up        up
Ethernet0/1    45.0.0.1        YES manual up        up
Ethernet0/2    unassigned      YES unset  administratively down down
Ethernet0/3    unassigned      YES unset  administratively down down

Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
S    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/0
L    24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/1
L    45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1

Router#
Router#
Router#

R3
Router#
Router#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    23.0.0.2        YES manual up        up
Ethernet0/1    32.0.0.1        YES manual up        up
Ethernet0/2    unassigned      YES unset  administratively down down
Ethernet0/3    unassigned      YES unset  administratively down down

Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
S    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 23.0.0.1
S    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1

Router#
Router#
Router#
```

R5,R6: Route Table

```
R5
R5#
R5#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    172.36.0.1      YES manual up        up
Ethernet0/1    32.0.0.2        YES manual up        up
Ethernet0/2    45.0.0.2        YES manual up        up
Ethernet0/3    unassigned      YES TFTP  down        down

R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 32.0.0.1
S    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
S    172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
S    223.0.50.0/24 [1/0] via 32.0.0.1

R6
R6#
R6#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0    223.0.50.1      YES manual up        up
Ethernet0/1    26.0.0.2        YES manual up        up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
S    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
S    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    223.0.50.0/24 is directly connected, Ethernet0/0
L    223.0.50.1/32 is directly connected, Ethernet0/0

R6#
R6#
R6#
```